Assignment

C Programming: Arrays & 2D Arrays

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Q1. Write a program to input n integers into an array and perform the following using loops:

(a) Find the largest and smallest number.

Output:

```
How many numbers you want to enter: 3
Enter 3 numbers:
1
2
3
The largest number is: 3
The smallest number is: 1
```

(b) Calculate the average of all numbers.

Output:

```
How many numbers you want to enter: 3
Enter 3 numbers:
5
79
189
The average of all numbers: 91.00
```

(c) Count how many numbers are even and how many are odd.

Output:

```
How many numbers you want to enter: 3
Enter 3 numbers:
7
8
6
Even numbers: 2
Odd numbers: 1
```

Q2. Write a program to input n integers in an array. Ask the user for a number and check if it exists in the array using the ternary operator. Print "Found" or "Not Found".

Output:

```
How many numbers you want to enter: 5
Enter 5 numbers:
78
5
6
43
56
Enter a number: 6
Found
```

```
How many numbers you want to enter: 5
Enter 5 numbers:
78
5
6
43
56
Enter a number: 50
Not Found
```

Q3. Write a program to input 10 integers (range 0–9). Store them in an array. Then, using a switch-case, print the frequency of each digit.

Output:

```
Enter 10 integers (between 0-9):
7
8
9
51
enter a number between (0-9):
6
7
9
20
enter a number between (0-9):
46
enter a number between (0-9):
5
8
9
6
0 - 0 times
1 - 0 times
2 - 0 times
3 - 0 times
4 - 0 times
5 - 1 times
6 - 2 times
7 - 2 times
8 - 2 times
9 - 3 times
```

Q4. Write a program to input two 3 × 3 matrices. Perform and display:

(a) Matrix addition

Output:

```
Enter 9 elements for the first matrix:
7
8
9
4
8
6
1
1
2
Enter 9 elements for the second matrix:
3
5
4
5
8
6
4

5
6

The sum of the two matrices is:
10      13      13
9       16      12
5       6       8
```

(b) Matrix subtraction

Output:

```
Enter 9 elements for the first matrix:
8
46
79
56
23
456
89

56
4888
Enter 9 elements for the second matrix:
89
4545
15
66
89
56
12
2
78

The difference of the two matrices is:
-81     -4499   64
-10     -66     400
77      54      4810
```

(c) Matrix multiplication

Output:

```
Enter 9 elements for the first matrix:
12
4
8
3
5
4
9
4
4
Enter 9 elements for the second matrix:
9
4
1
2
6
0
6
8
5

The product of the two matrices is:
48      64      40
24      32      20
24      32      20
```

Q5. Write a program to find the transpose of a given 3 × 3 matrix

Output:

```
Enter 9 elements for matrix:
1
2
3
1
2
3
1
2
3
transpose of matrix is:
1       1       1
2       2       2
3       3       3
```

Q6. Write a program that takes a 4 × 4 matrix and finds the maximum element in each row and each column.

Output:

```
Enter 16 elements for 4x4 matrix:
1
2
5
6
8
8
4
3
56
7
9
4
32
5

89
7

Max in each Row
Maximum in Row 0 is: 6
Maximum in Row 1 is: 8
Maximum in Row 2 is: 56
Maximum in Row 3 is: 89

Max in each Column
Maximum in Column 0 is: 56
Maximum in Column 1 is: 8
Maximum in Column 2 is: 89
Maximum in Column 3 is: 7
```

Q7. Assume a class of 5 students, each having marks in 3 subjects. Store the marks in a 2D array where each row represents a student and each column represents a subject.
Perform the following:

(a) Calculate the total and average marks of each student.

Output:

```
Enter maximum marks for subject: 100
Enter marks for Student 1 (out of 100):
Subject 1: 56
Subject 2: 87
Subject 3: 123
Enter a value between 0 and 100.
Subject 3: 45

Enter marks for Student 2 (out of 100):
Subject 1: -12
Enter a value between 0 and 100.
Subject 1: 76
Subject 2: 56
Subject 3: 94

Enter marks for Student 3 (out of 100):
Subject 1: 23
Subject 2: 56
Subject 3: 79

Enter marks for Student 4 (out of 100):
Subject 1: 95
Subject 2: 67
Subject 3: 39

Enter marks for Student 5 (out of 100):
Subject 1: 91
Subject 2: 84
Subject 3: 76
```

```
Result
Student 1:
Total Marks: 188/300
Average:  62.67

Student 2:
Total Marks: 226/300
Average:  75.33

Student 3:
Total Marks: 158/300
Average:  52.67

Student 4:
Total Marks: 201/300
Average:  67.00

Student 5:
Total Marks: 251/300
Average:  83.67
```

(b) Find the highest scorer (student with maximum total marks).

Output:

```
Enter maximum marks for a subject: 100

Enter marks for Student 1 (out of 100):
Subject 1: 46
Subject 2: 79
Subject 3: 86
Enter marks for Student 2 (out of 100):
Subject 1: 26
Subject 2: 53
Subject 3: 45
Enter marks for Student 3 (out of 100):
Subject 1: 62
Subject 2: 65
Subject 3: 49
Enter marks for Student 4 (out of 100):
Subject 1: 56
Subject 2: 23
Subject 3: 45
Enter marks for Student 5 (out of 100):
Subject 1: 56
Subject 2: 68
Subject 3: 98
Student 5 is the top scorer with 222 marks.
```

(c) Find the subject in which the class performed worst (lowest average marks).

Output:

```
Enter maximum marks for a subject: 100

Enter marks for student 1 (out of 100):
Subject 1: 45
Subject 2: 79
Subject 3: 56
Enter marks for student 2 (out of 100):
Subject 1: 15
Subject 2: 23
Subject 3: 15
Enter marks for student 3 (out of 100):
Subject 1: 46
Subject 2: 56
Subject 3: 46
Enter marks for student 4 (out of 100):
Subject 1: 56
Subject 2: 2
Subject 3: 79
Enter marks for student 5 (out of 100):
Subject 1: 53
Subject 2: 45
Subject 3: 79
class performed worst in subject 2.
```

(d) Use sizeof() operator to display the total memory consumed by the array.

Output:

```
Total memory consumed by the marks array is 60 bytes.


...Program finished with exit code 0
Press ENTER to exit console.
```